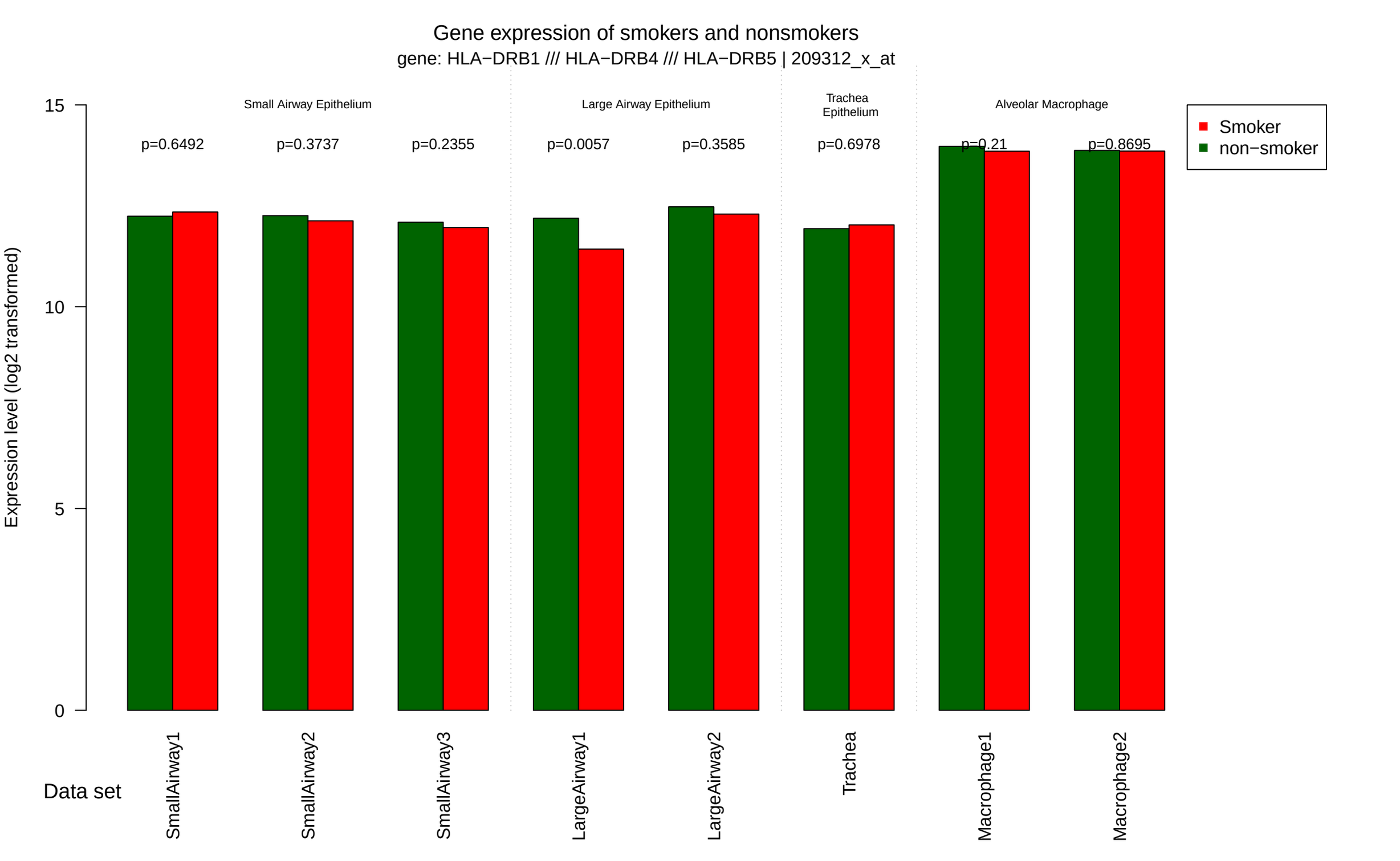
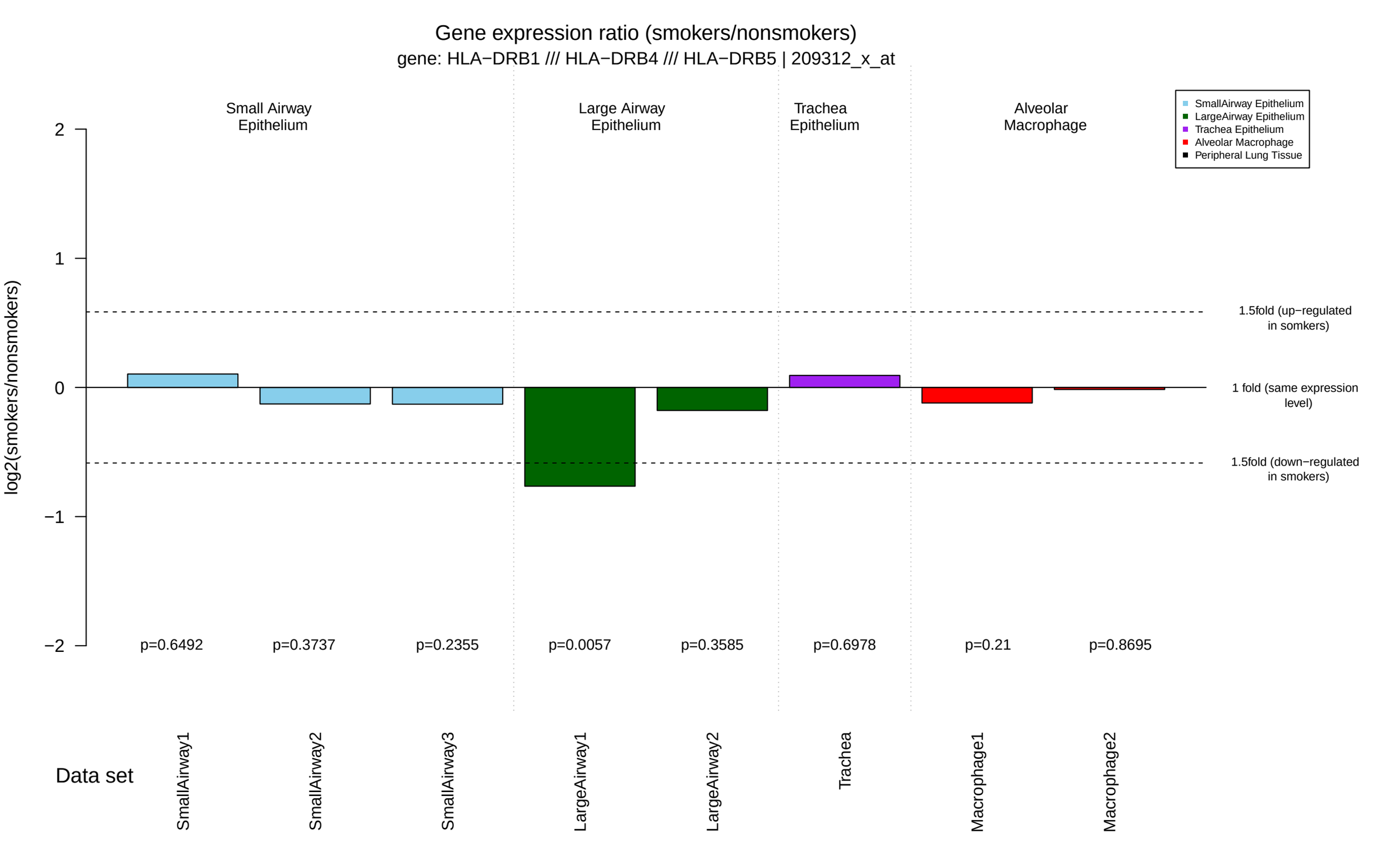
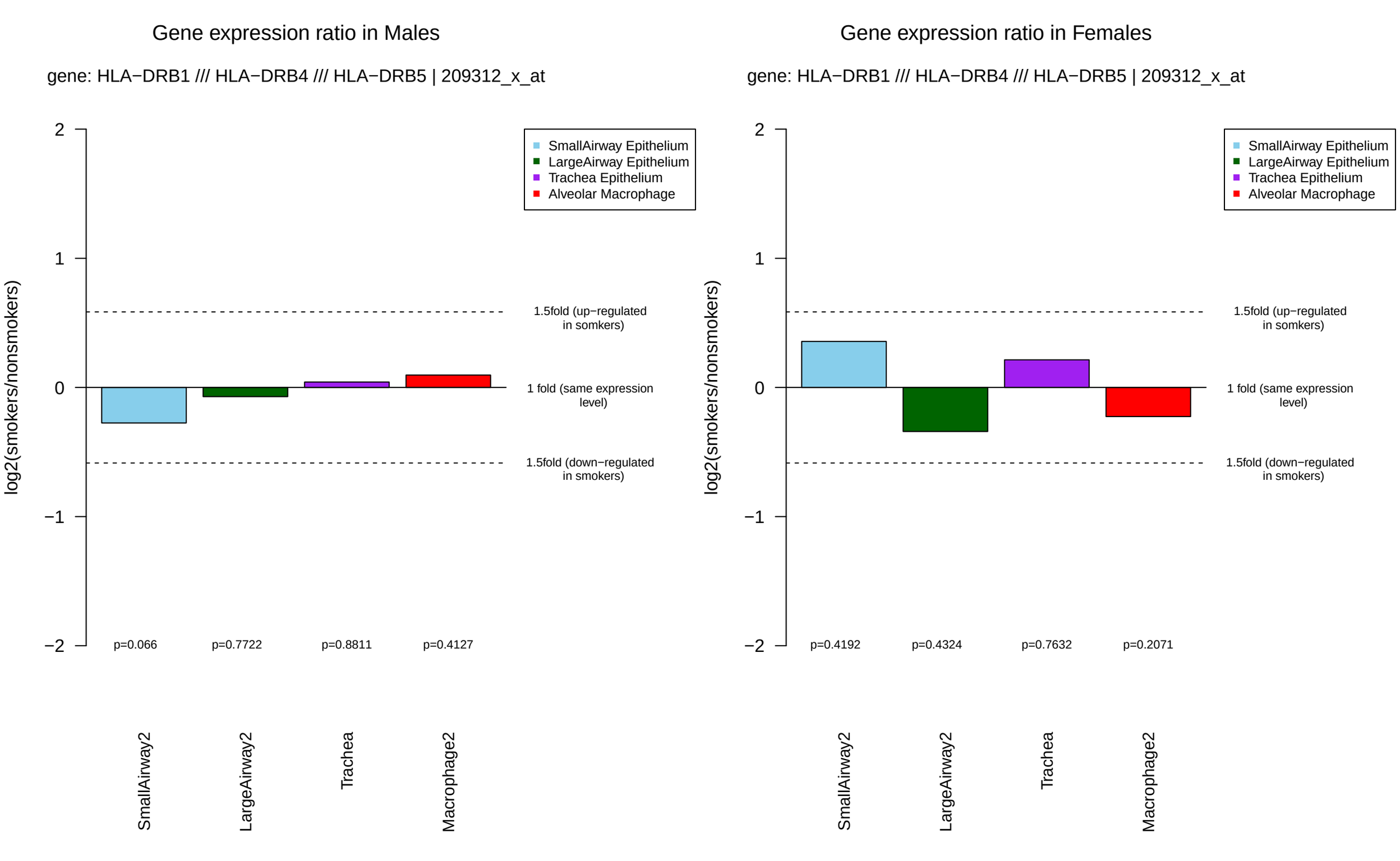
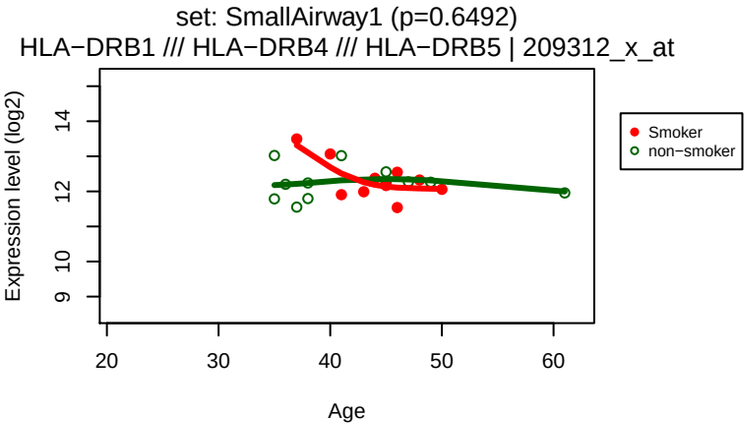
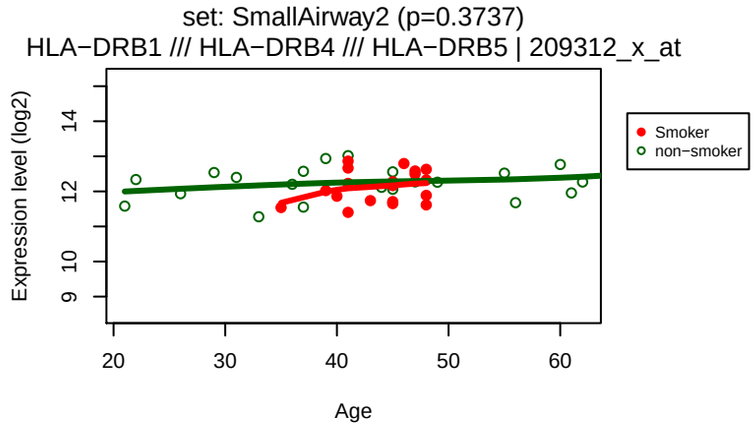
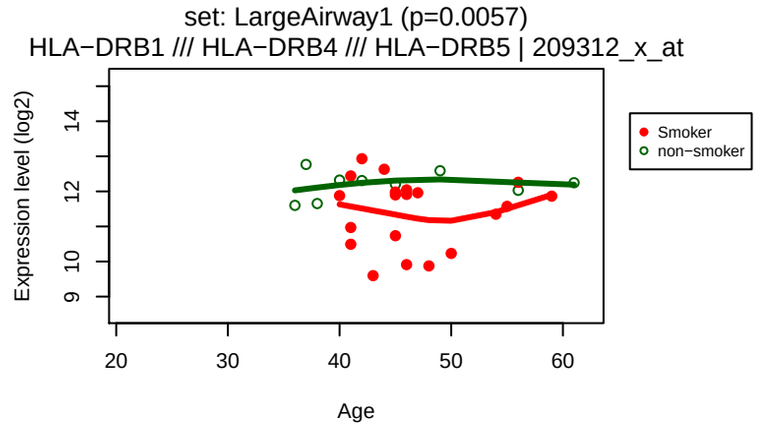

Gene expression of smokers and nonsmokers
gene: HLA−DRB1 /// HLA−DRB4 /// HLA−DRB5 | 209312_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6492
p=0.3737
p=0.2355
p=0.0057
p=0.3585
p=0.6978
p=0.21
p=0.8695
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

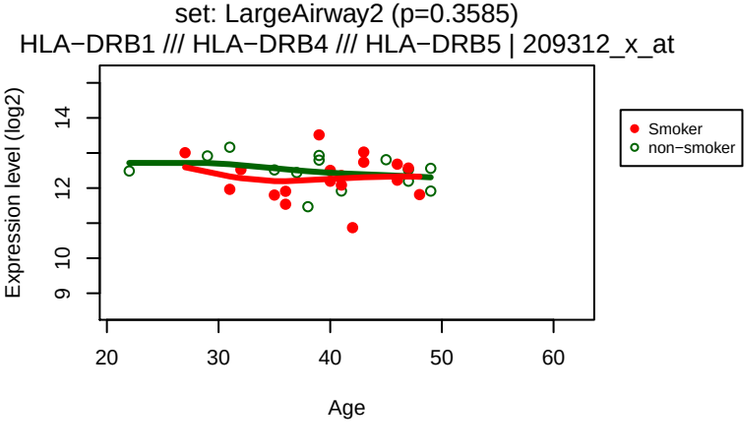

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DRB1 /// HLA−DRB4 /// HLA−DRB5 | 209312_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6492
p=0.3737
p=0.2355
p=0.0057
p=0.3585
p=0.6978
p=0.21
p=0.8695
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

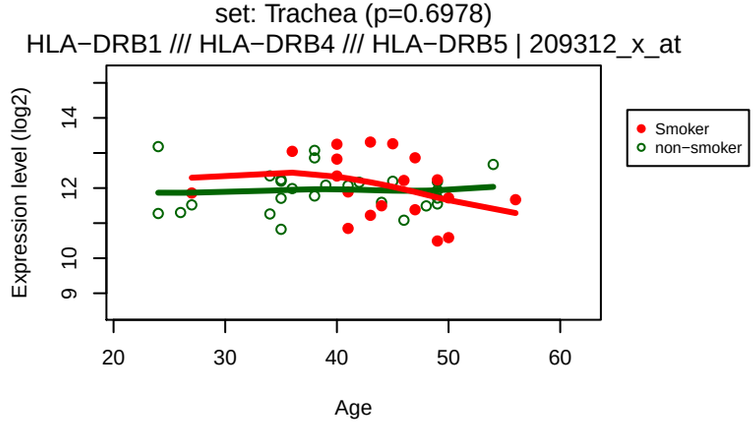

Gene expression ratio in Males
gene: HLA−DRB1 /// HLA−DRB4 /// HLA−DRB5 | 209312_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.066
p=0.7722
p=0.8811
p=0.4127
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DRB1 /// HLA−DRB4 /// HLA−DRB5 | 209312_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4192
p=0.4324
p=0.7632
p=0.2071
SmallAirway2
LargeAirway2
Trachea
Macrophage2

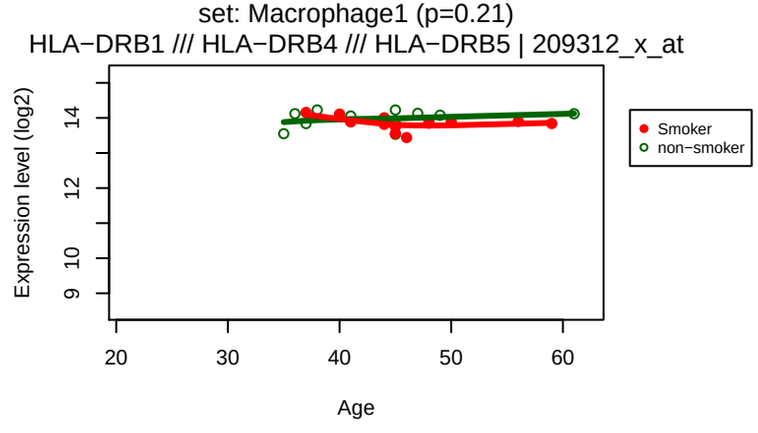

set: SmallAirway1 (p=0.6492)
HLA−DRB1 /// HLA−DRB4 /// HLA−DRB5 | 209312_x_at
Expression level (log2)
Age
Smoker
non−smoker

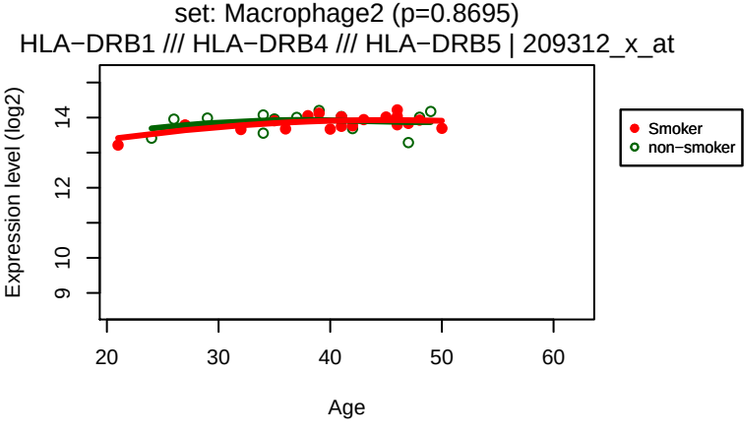

set: SmallAirway2 (p=0.3737)
HLA−DRB1 /// HLA−DRB4 /// HLA−DRB5 | 209312_x_at
Expression level (log2)
Age
Smoker
non−smoker

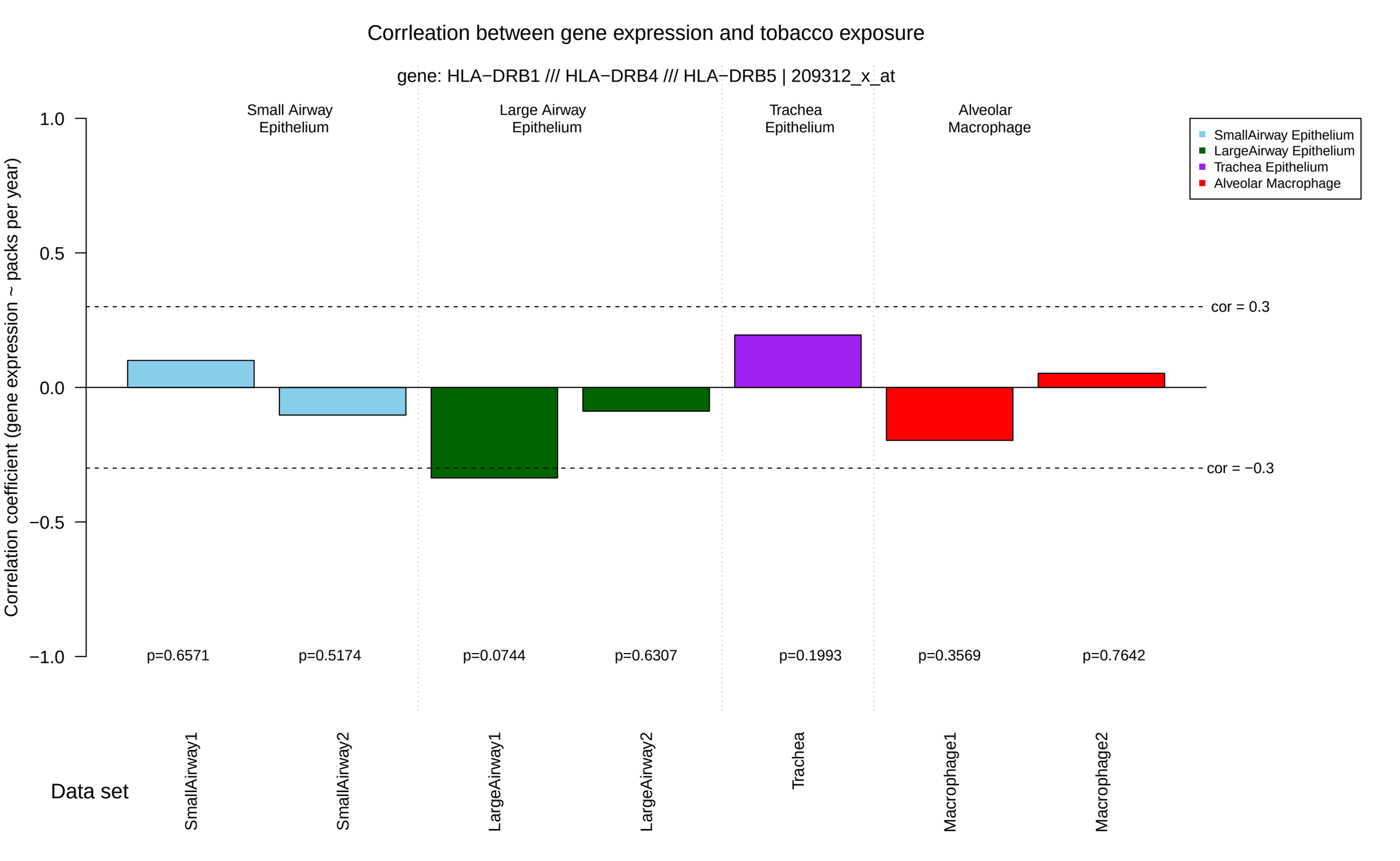

set: LargeAirway1 (p=0.0057)
HLA−DRB1 /// HLA−DRB4 /// HLA−DRB5 | 209312_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3585)
HLA−DRB1 /// HLA−DRB4 /// HLA−DRB5 | 209312_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6978)
HLA−DRB1 /// HLA−DRB4 /// HLA−DRB5 | 209312_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.21)
HLA−DRB1 /// HLA−DRB4 /// HLA−DRB5 | 209312_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8695)
HLA−DRB1 /// HLA−DRB4 /// HLA−DRB5 | 209312_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DRB1 /// HLA−DRB4 /// HLA−DRB5 | 209312_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6571
p=0.5174
p=0.0744
p=0.6307
p=0.1993
p=0.3569
p=0.7642
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2